Name_____ Date _____

# Measuring Rate

## Key Ideas

**Fill in the blanks with the term or terms that makes each statement true.**

A _____ is a comparison of the size of one number as compared to the size of another number.

Two ratios are said to be _____ if they have the same value.

A ratio that compares two different units is called a(n) _____.

A rate that compares a quantity per *one* of the other quantity is called a(n) _____.

| Word Bank |
| --- |
| **unit rate** |
| **ratio** |
| **rate** |
| **equivalent** |

## Review

**Determine which equations show a true statement. If the fractions are equivalent, put a "✓" next to them. If they are not equivalent, put an "X" through them.**

a. $\frac{1}{2} = \frac{6}{12}$ b. $\frac{4}{5} = \frac{12}{20}$ c. $\frac{4}{18} = \frac{2}{9}$ d. $\frac{6}{10} = \frac{15}{25}$

## Guided Practice

**Follow the example to find the unit rate for each rate shown below.**

**Rate:**
a ratio comparing two different units

division sign

**12 miles / 3 hours**

we read this as "12 miles *per* 3 hours"

**Unit Rate:**
the first quantity per *one* of the second quantity

**12 ÷ 3 = 4**

**4 miles / 1 hour or 4 miles/hour**

a. 9 gallons / 3 seconds

b. $40.00 / 4 hours

c. 40 heartbeats / 30 seconds

d. 100 steps / 2 minutes

©Taylor J's MATH materials

## Practice

**1. Write down the rate in each scenario. Then, find the unit rate.**

a. buying 12 pencils for $3.00

b. traveling 80 miles in 2 hours

c. walking 2 miles in 24 minutes

d. peeling 8 potatoes in 5 minutes

**2. Are the rates and unit rates in each scenario above considered to be equivalent? Why or why not?**

**3. Determine which pairs of ratios below are equivalent. If the ratios are equivalent, put a "✓" next to them. If they are not equivalent, put an "X" through them.**

a. 5 to 3, 20 to 12

b. $\frac{14}{22}$, $\frac{7}{11}$

c. 6:4, 24:8

**4. For each ratio below, write an equivalent ratio.**

a. 18 to 30

b. 50:200

c. $\frac{2}{3}$

d. 60 to 12

e. $\frac{1}{4}$

f. 70:30

g. 32:8

h. 350 to 500

i. $\frac{40}{10}$

©Taylor J's MATH materials

**5. Seth bakes 100 cookies in 2 hours. Erica bakes 200 cookies in 2.5 hours.**

a. Write the rate at which each person is baking cookies.

b. Write the unit rate at which each person is baking cookies.

c. Which person is baking at a faster rate? How you know?

**6. Brandon can swim 15 laps around the pool in 40 minutes. Peter can swim 10 laps in 25 minutes.**

a. Write the rate at which each boy swims.

b. Write the unit rate at which each boy swims.

c. If both boys continue swimming, which boy will swim 20 laps first? How do you know?

**7. Jasmine buys a pack of 15 socks for $9.00. Shawn buys a pack of 20 socks for $11.00. Write a ratio to represent they paid to the number of socks. Who got a better deal on socks? Explain.**

**8. Mario and his 3 friends attend a professional football game. After parking, tickets, and food, they spent $240.00 total. Mario tells his friends that he got a better deal last week, paying only $50/person. How much better of a deal did Mario get last week?**

©Taylor J's MATH materials